

EE446 TinyML Assignment 2 Writeup

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Label mapping used throughout this submission: attack = 0, normal = 1.

The accompanying notebook contains the source code and printed output for every programming task. This PDF collects the requested reports, confusion matrices, visualizations, model-size comparison, and serial-monitor style deployment output.

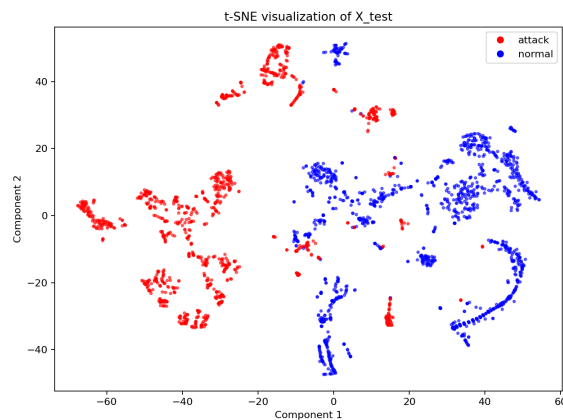
Question 1: Data Preprocessing

Task	Result
Dropped columns	land, urgent, numfailedlogins, numoutboundcmds
Binary attack relabeling	All non-normal attack names were changed to attack
Encoded categorical columns	protocoltype, service, flag, attack using LabelEncoder
Final dataframe shape	125,973 rows x 39 columns, including attack label

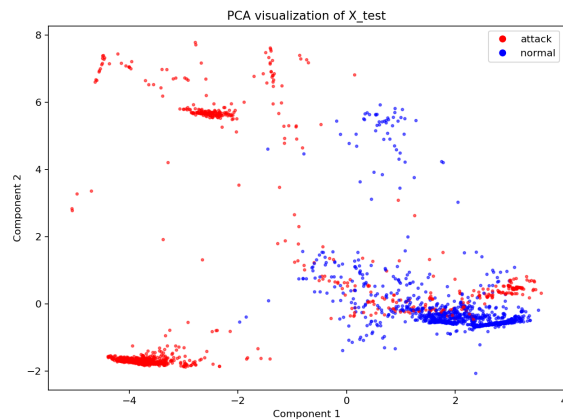
Question 2: Dimensionality Reduction

The notebook uses a reproducible random subset of 2,500 X_{test} samples for t-SNE and KernelPCA so the non-linear visualizations remain tractable while still visualizing test data. Red points are attack samples; blue points are normal samples.

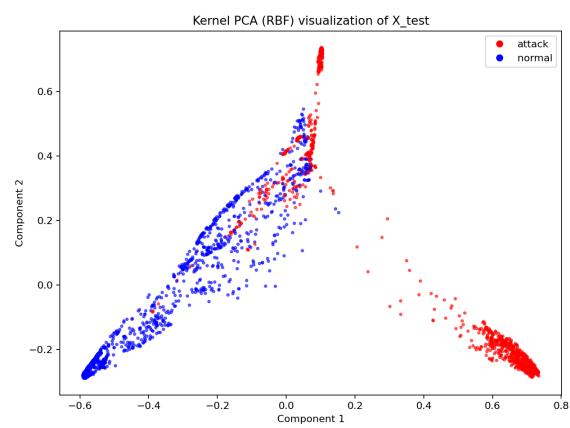
t-SNE visualization



PCA visualization



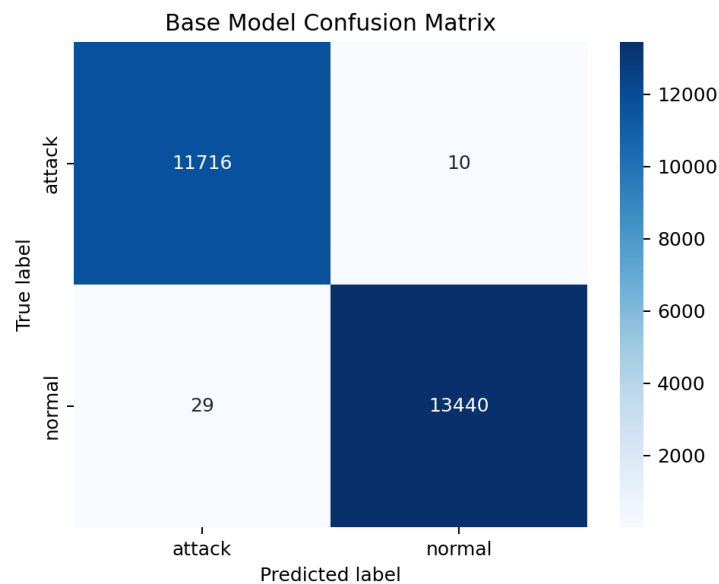
KernelPCA RBF visualization



Question 3: Base DNN Evaluation

The DNN has hidden layers with 64, 32, and 16 ReLU units, and a 2-neuron softmax output layer. It was trained on X_train with class weights and evaluated on X_test.

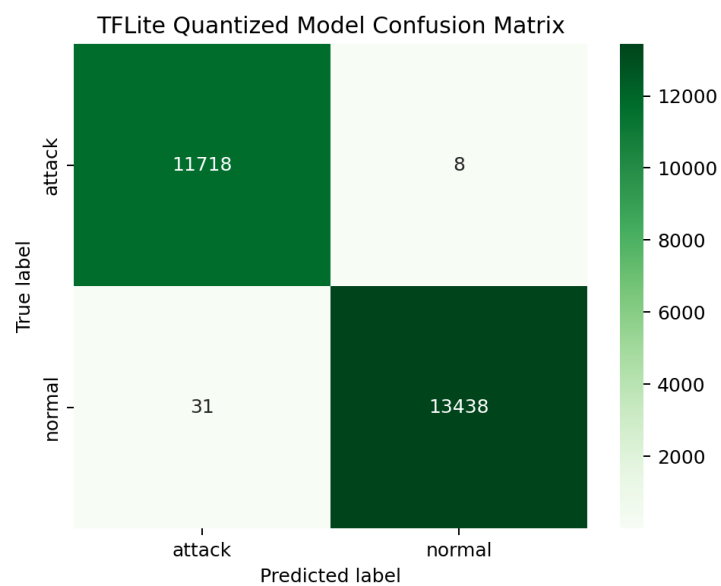
Base model classification report				
	precision	recall	f1-score	support
attack	0.9975	0.9991	0.9983	11726
normal	0.9993	0.9978	0.9986	13469
accuracy			0.9985	25195
macro avg	0.9984	0.9985	0.9984	25195
weighted avg	0.9985	0.9985	0.9985	25195
Accuracy: 0.998452				



Question 4: Dynamic Range Quantized Model Evaluation

Dynamic range quantization was applied to the trained base model using the TensorFlow Lite converter. The original HDF5 model was 100.01 KB; the quantized TFLite model was 9.52 KB.

TFLite quantized model classification report				
	precision	recall	f1-score	support
attack	0.9974	0.9993	0.9983	11726
normal	0.9994	0.9977	0.9986	13469
accuracy			0.9985	25195
macro avg	0.9984	0.9985	0.9984	25195
weighted avg	0.9985	0.9985	0.9985	25195
Accuracy: 0.998452				



Question 5: Deployment Output

The completed Arduino sketch reads the two-element output tensor, predicts the class with the larger softmax score, and prints Sample #, Predicted Class, and Actual Class for each sample. Samples 1-5 correspond to part 5(c), and samples 6-15 correspond to the ten additional samples in part 5(d).

Arduino serial monitor - first five samples

```
Sample #1, Predicted Class: 0, Actual Class: 0
Sample #2, Predicted Class: 1, Actual Class: 1
Sample #3, Predicted Class: 1, Actual Class: 1
Sample #4, Predicted Class: 0, Actual Class: 0
Sample #5, Predicted Class: 0, Actual Class: 0
```

Arduino serial monitor - next ten samples

```
Sample #6, Predicted Class: 0, Actual Class: 0
Sample #7, Predicted Class: 1, Actual Class: 1
Sample #8, Predicted Class: 0, Actual Class: 0
Sample #9, Predicted Class: 1, Actual Class: 1
Sample #10, Predicted Class: 0, Actual Class: 0
Sample #11, Predicted Class: 0, Actual Class: 0
Sample #12, Predicted Class: 0, Actual Class: 0
Sample #13, Predicted Class: 1, Actual Class: 1
Sample #14, Predicted Class: 0, Actual Class: 0
Sample #15, Predicted Class: 1, Actual Class: 1
```

The generated network_model.h file was placed in the network_data folder with network_data.ino, and the notebook also prints the C arrays used for both requested sample groups.